DSP – LEC – 9 - Ch- 4- Obj.

CHAPTER – 4

Z-TRANSFORM AND ITS APPLICATIONS IN DSP

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Z-TRANSFORM AND ITS APPL.s IN DSP

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4.1 Introduction

4.2 The Z-transform

4.3 The inverse Z-transform

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Sec. 4.1

4.1 INTRODUCTION

- It is a convenient yet invaluable tool for representing, analog signal, and designing discrete time signals and system.

- It plays a similar role in discrete-time system to that which Laplace transform plays in continuous time system.

- A discrete time system is essentially a mathematical algorithms that takes an i/p seq., **x(n)**, and produces an output sequence, **y(n)**.

Sec. 4.1 Cont. - 1

- **Examples of D-T systems are:**
  - * Digital filters,
  - * digital controller and
  - * digital spectrum analyzer.

A discrete Time systems may be:

* Linear or non-linear systems.

* Time invariant or time varying system.

* Causal or a non-causal system.

* Stable or un-stable system.

Sec. 4.1 Cont. - 2

*** Linear Systems:**

- A discrete-time system is said to be linear if it obeys the principles of superposition,

i.e., the response of a linear system to two or more **i/ps** is equal to the sum of the responses of the system to all the other i/ps.

Sec. 4.1 Cont. - 3

- **For example:**

  if for input say $\mathbf{x_1(n)}$ the system gives the output $y_1(n)$

  and

  for input, say, $\mathbf{x_2(n)}$ the system gives the output $\mathbf{y_2(n)}$.

- The response of the system to both inputs will be:

$$a_1 x_1(n) + a_2 x_2(n) \longrightarrow a_1 y_1(n) + a_2 y_2(n)$$

where

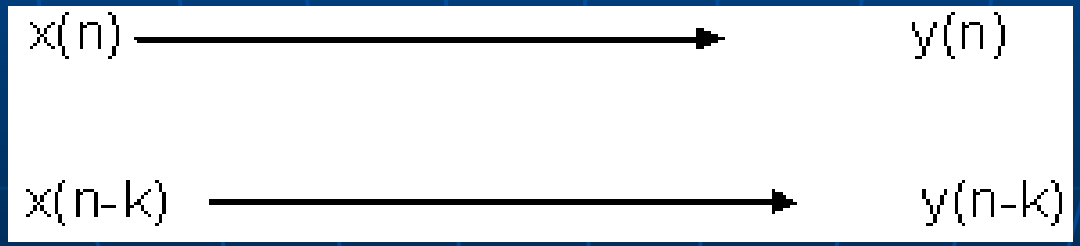
a_1 a_2 **....** are arbitrary constants.

Sec. 4.1 Cont. - 4

* Time Invariant systems

- A discrete-time system is said to be time invariant (Some times referred to as shift invariant) if its output is independent of the time the input is applied.

e.g., if the input is **x(n)**, the system o/p is **y(n)** and for **x(n-k)** input, the system o/p is **y(n-k)** as,



i.e., delay in the i/p causes delay by the same amount in the o/p.

Sec. 4.1 Cont. - 5

*** Causal Systems:**

- The input relationship of an LTIS is given by the convolution sum, given by:

$$y(n) = \sum_{k=-\infty}^{\infty} h(k)\, x(n-k)$$

Where

h(k) ….. is the impulse response of the system.

The values of **h(k)** completely define the DTS in the time domain.

Sec. 4.1 Cont. - 6

- The system is used to be causal system if it produces an o/p only where there is an input.

- **Note that**, all physical systems are causal.

- In general, a causal D-T-System has zero impulse response before time zero. **i.e**.

$$h(k)=0 \quad \text{for all } k<0$$

Sec. 4.1 Cont. - 7

- Also the D-T-sequence, say, x(n) is causal if its value is zero before time zero. **i.e**.

$$x(n)=0 \quad \text{for all } n<0$$

- For continuous signals or system, they are causal if:

$$x(t) = 0 \quad \text{for all } t<0 \quad \text{for signal.}$$

$$h(t) = 0 \quad \text{for all } t<0 \quad \text{for system.}$$

Sec. 4.1 Cont. - 8

*** System Stability:**

- LTIS is stable if its impulse response h(k), satisfied the following condition:

$$\sum_{k=-\infty}^{\infty} |h(k)| < \infty$$

This condition is satisfied if , **h(k)** is of finite duration. **Or**, if h(k) decays towards zero as k increase.

Sec. 4.2

4.2 Z - TRANSFORM

z-trans. of a seq., **x(n)**, which is valid for all **"n"**, is defined as:

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n} \qquad (4.1)$$

Where:

z …… is complex variable.

Sec. 4.2 Cont. - 1

In causal systems, x(n) may be nonzero only in the interval $0 < n < \infty$ and **Eq. 4.1** reduces to the so – called one-sided z-transform:

$$X(z) = \sum_{n=0}^{\infty} x(n) z^{-n} \qquad (4.2)$$

Sec. 4.2 Cont. - 2

Region of convergence (ROC)

- The region where the z-transform converges is known as the region of convergence (ROC), and in this region the values of X(z) are finite.

Sec. 4.2 Cont. - 3

Example 4.1

Find the z-transform and the ROC for each of the discrete-time sequence given in **Fig. 4.1.**

Sec. 4.2

Example 4.1 - Cont - 1

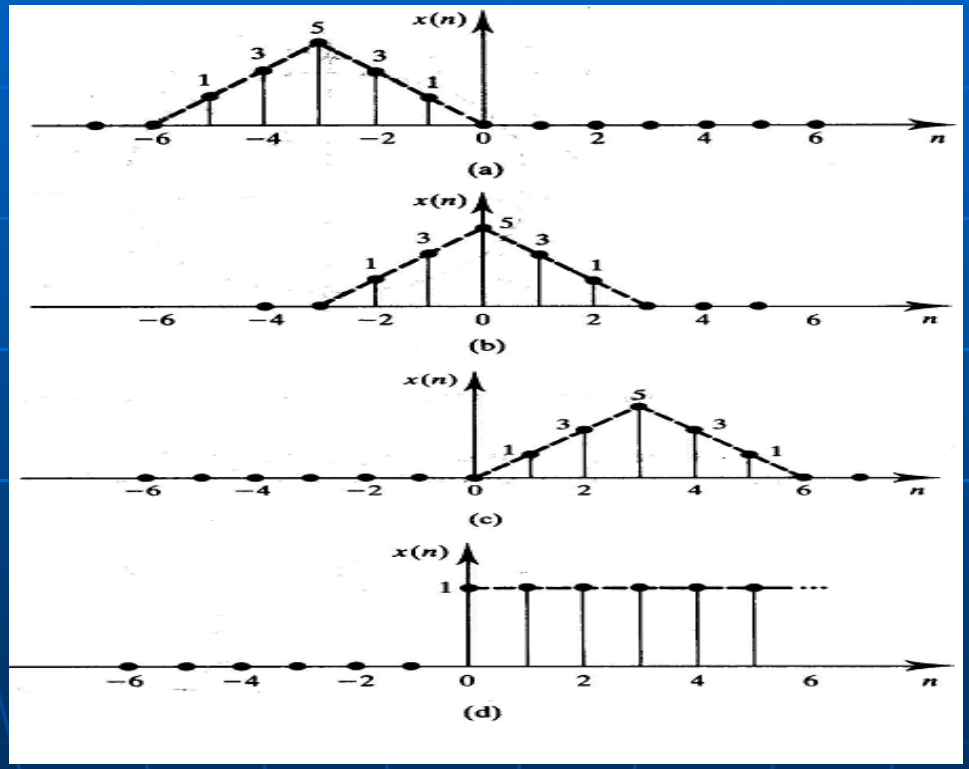


Fig. 4.1 Causal and non-causal discrete-time seq.s.

Sec. 4.2

Example 4.1 - Cont - 2

(a) The sequence of Fig. 4.1(a) is:

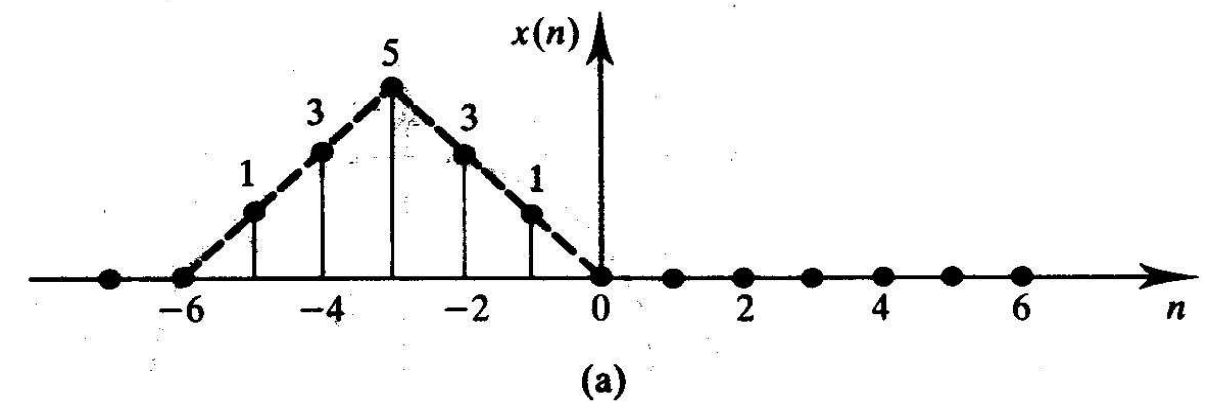


Fig. 4.1(a)

Example 4.1 - Cont - 3

- **The sequence of Fig. 4.1(a) is defined as:**

  - non-causal,
  - it is of a finite duration.
  - The sequence has values:

    x(-6) = x(0) = 0,
    x(-5) = x(-1)= 1,
    x(-4) = x(-2)= 3, and
    x(-3) = 5,

Sec. 4.2

Example 4.1 - Cont - 4

From **Eq. 4.1**, the z-transform is given by:

$$X_1(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n} = z^5 + 3z^4 + 5z^3 + 3z^2 + z$$

It is clear that the value of X(z) becomes infinite when $z = \infty$.

Thus ROC is everywhere in the z-plane except at $z = \infty$.

Example 4.1 - Cont - 5

(b) The sequence of Fig. 4.1(b) is:

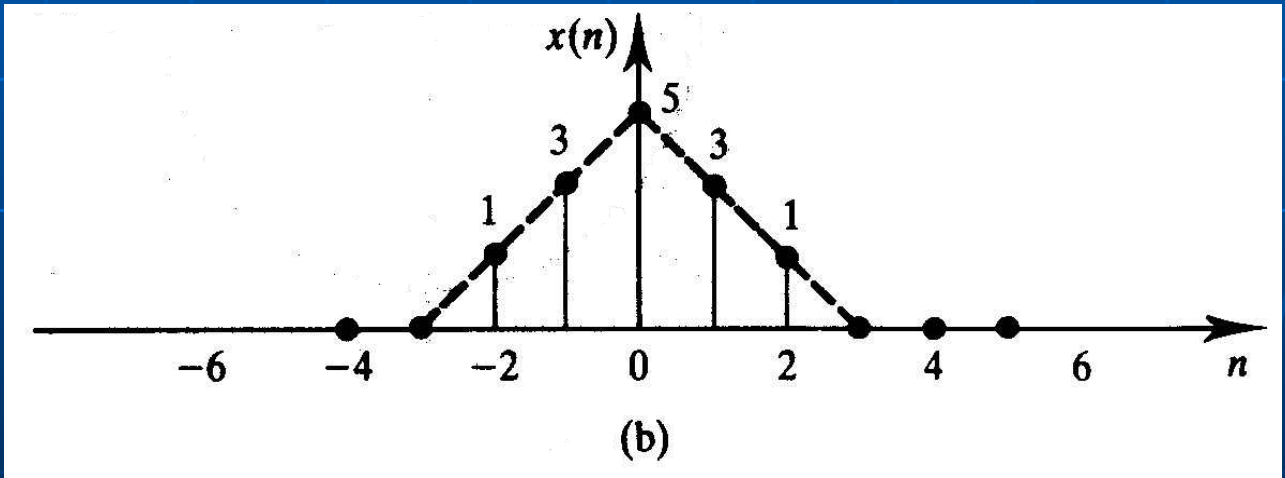


Fig. 4.1(b)

Sec. 4.2

Example 4.1 - Cont - 6

Again, the sequence in Fig. 4.1(b) is not causal. It is of a finite duration, and double sided. The values of the sequence are x(-3) =0, x(-2) = 1, x(-1) = 3 , x(0) = 5, x(1) = 3, x(2) = 1 and x(3) = 0. From Equation 3.1, the z-transform is given by

$$X_2(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n} = z^2 + 3z + 5 + 3z^{-1} + z^{-2}$$

It is evident that the value of X(z) is infinite if **z =0** or if **z = ∞** . Therefore, the ROC is everywhere except at Z = 0 and Z = ∞ .

Example 4.1 - Cont - 7

(c) The sequence of Fig. 4.1(c) is:

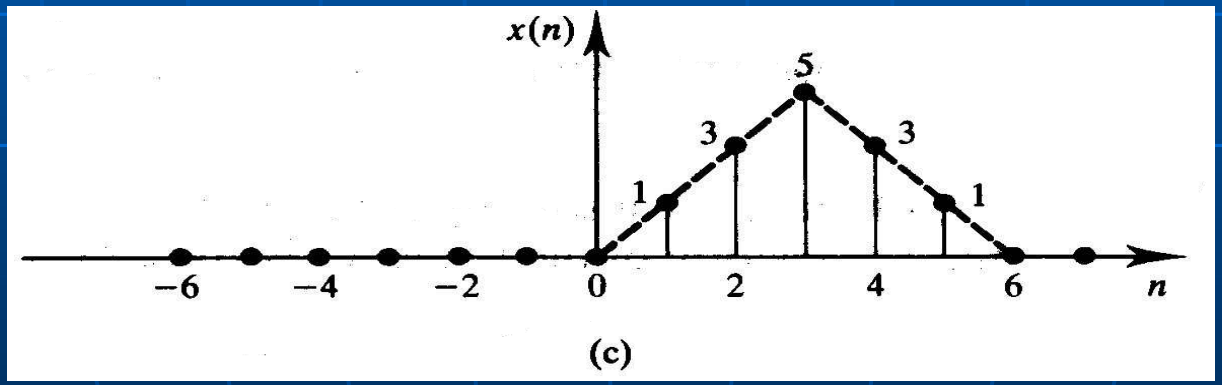


Fig. 4.1(c)

Sec. 4.2

Example 4.1 - Cont - 8

Figure 4.1(c) represents a causal, finite duration sequence with values x(0) = 0, x(1) = 1, x(2) =3, x(3) = 5, x(4) = 3, x(5) = 1 and x(6) =0. The z-transform is given by

$$X_3(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n} = z^{-1} + 3z^{-2} + 5z^{-3} + 3z^{-4} + z^{-5}$$

In this case, X(z) = ∞ for z = 0. Thus the region of convergence is every where except at z = 0.

Example 4.1 - Cont - 9

(d) The discrete-time sequence in Fig. 4.1(d): may be defined mathematically as:

$$x(n) = 1 \qquad 0 \le n \le \infty$$
$$= 0 \qquad n < 0$$

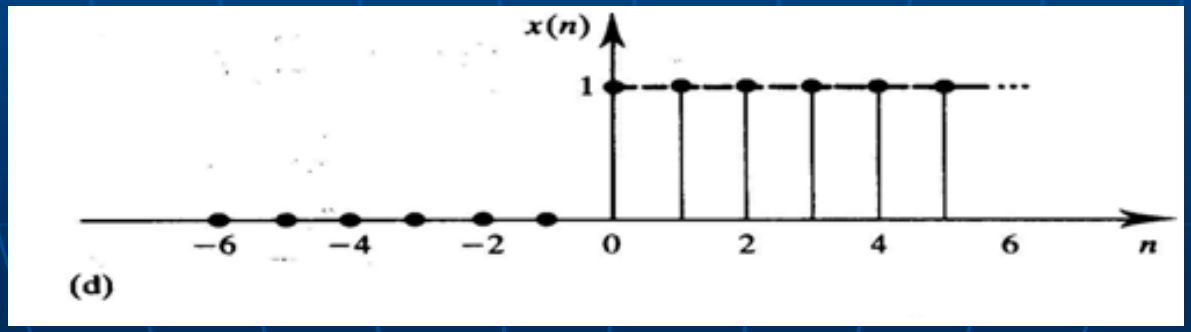


Fig. 4.1d.

Example 4.1 - Cont - 10

The sequence of Fig. 4.1(d) is:

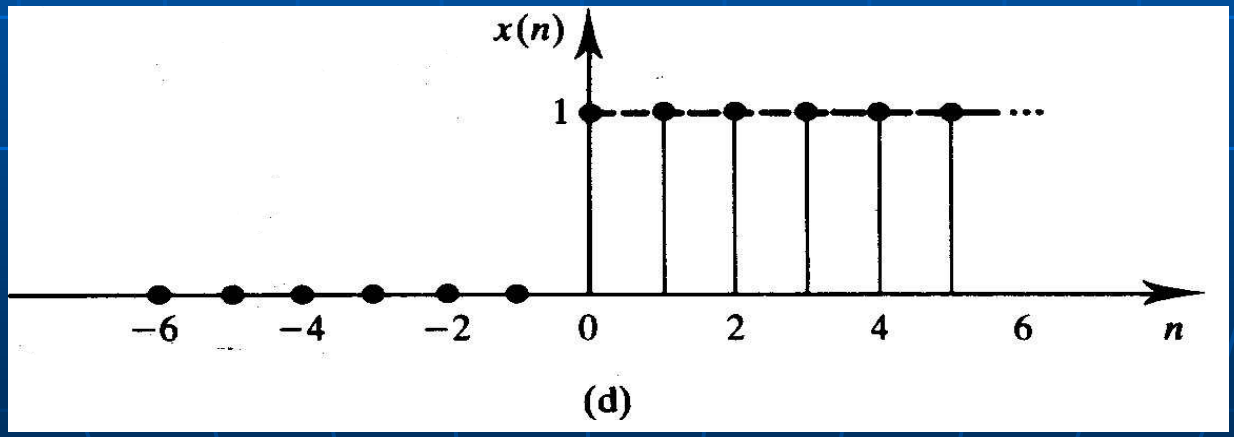


Fig. 4.1(d)

Example 4.1 - Cont - 11

The discrete-time sequence in **Fig. 4.1(d):**

- it is a causal sequence,
- It is of infinite duration.
- From Eq. 4.1, the z-transform of the sequence is given by:

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

$$= \sum_{n=0}^{\infty} z^{-n}$$

$$= 1 + z^{-1} + z^{-2} + \ldots\ldots$$

Example 4.1 - Cont - 12

This is a geometric series with a common ratio of z^{-1}. The series converges if $|z^{-1}| < 1$ or equivalently if $|z| > 1$. Thus we may express X(z) in closed form provided that $|z| > 1$:

$$X(z) = 1 + z^{-1} + z^{-2} + \ldots$$
$$= 1/(1-z^{-1})$$
$$= z/(z-1) \qquad (4.3)$$

Example 4.1 - Cont - 13

- In this case, the Z-transform is valid everywhere outside a circle of unit radius whose center is at the origin.

- The exterior of the circle is the region of convergence (Fig. 4.2).

- We can readily verify that when $|z| > 1$ $X(z)$ converges whereas when $|z| < 1$ $X(z)$ diverges.

Example 4.1 - Cont - 14

- For example, if we let **z = 2** (outside the unit circle) we find that the series on the RHS of **Eq. 4.3** adds up to 2:

$$X(z) = 1 + 1/2 + (1/2)^2 + (1/2)^3 + \ldots = 2/(2-1) = 2$$

as it is clearly a geometric series with a common ratio of 1/2 and a first term of 1, giving the sum to infinity of 2/(2 – 1) = 2.

Example 4.1 - Cont - 15

On the other hand if z = 1/2 (inside the unit circle) the series of Eq. 4.3 becomes:

$$X(z) = 1 + 1/0.5 + (1/0.5)^2 + (1/0.5)^3 + (1/0.5)^4 + \ldots.$$
$$= 1 + 2 + 4 + 8 + \ldots\ldots$$

which is seen to be diverging.

Sec. 4.2

Example 4.1 - Cont - 16

- In Fig.4.2, the region of convergence (hatched) is seen to be bounded by the circle |z| =1, the radius of the pole of X(z).

- Values of z for which X(z) = are referred to as poles of X(z). Values of z for which X(z) = 0 are referred to as the zeros of X(z).

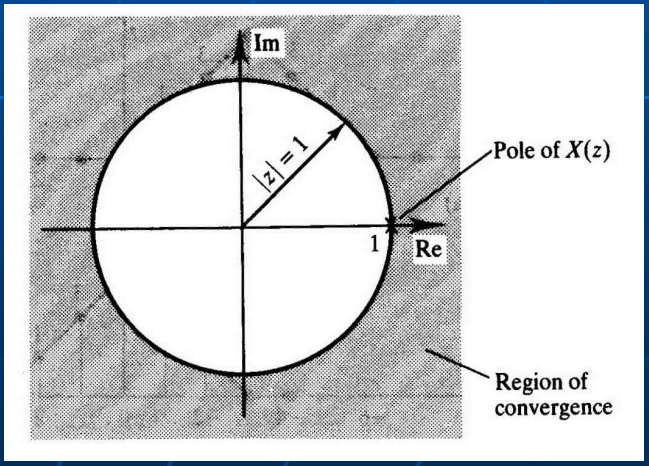


Fig. 4.2 Region of convergence for Example 4.1, part (4)

Example 4.1 - Cont - 17

From the above examples, we reach to:

- For causal sequences of finite duration the z-transform converges everywhere except at z = 0.

- For causal infinite duration sequences the z-transform converges everywhere outside a circle bounded by the radius of the pole with the largest radius.

- For stable causal systems the ROC always encloses the circle of unit radius which is important for the systems to have a frequency response.

Example 4.1 - Cont - 18

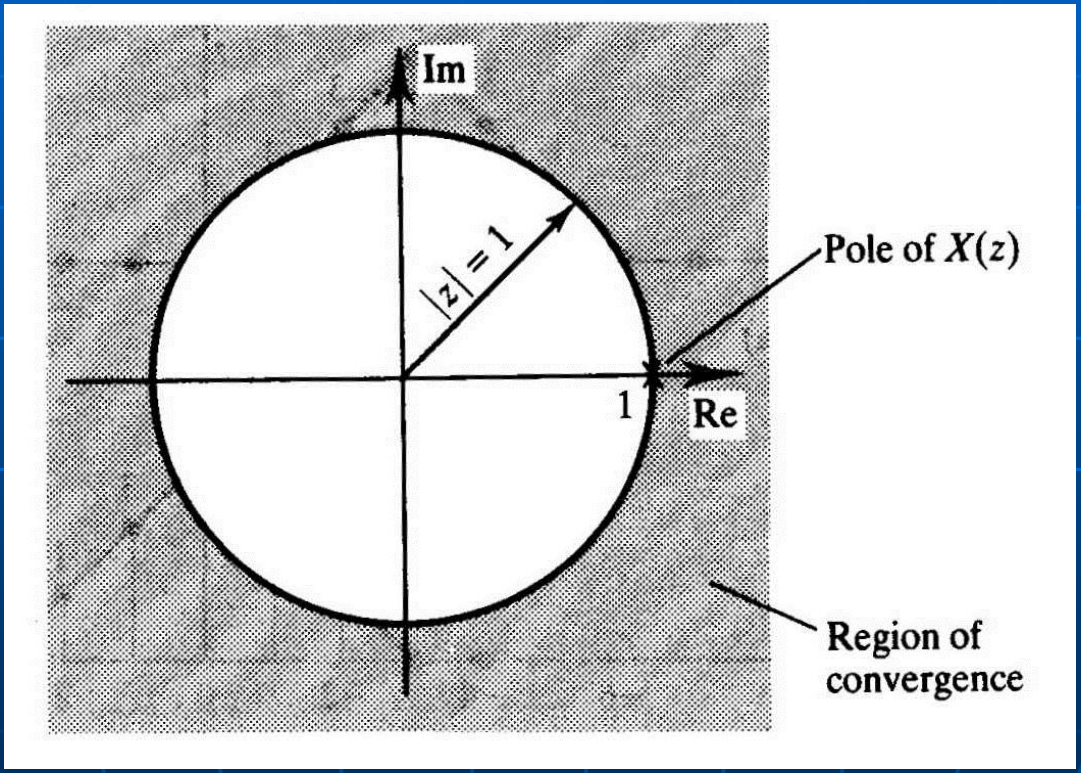


Fig. 4.2 Region of convergence for Example 4.1, part (4)

Sec. 4.3

4.3 INVERSE Z - TRANSFORM (IZT)

- IZT allows us to recover the discrete-time sequence, x(n), given its z-transform.

- IZT is particularly useful in DSP work, **e.g.,** in finding the impulse response of digital filters.

Sec. 4.3 Cont. - 1

- Symbolically, the **IZT** may be defined as:

$$x(n) = Z^{-1}[X(z)] \qquad (4.4)$$

Where:

X(z) …. is the z-transform of x(n) and

Z^{-1} … is the symbol for the IZT.

Sec. 4.3 Cont. - 2

- For a causal sequence, X(z), in Eq. 4.2 can be expanded into a power series as:

$$X(z) = \sum_{n=0}^{\infty} x(n)z^{-n}$$

$$= x(0) + x(1)z^{-1} + x(2)z^{-2} + x(3)z^{-3} + \ldots \qquad (4.5)$$

- It is seen that the values of **x(n)** are the coefficients of **z^{-n} (n=0,1,….):**

Sec. 4.3 Cont. - 3

- In practice, X(z) is often expressed as a ratio of two polynomials in z–1 or equivalently in z :

$$X(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2} + \ldots + a_N z^{-N}}{b_0 + b_1 z^{-1} + b_2 z^{-2} + \ldots + b_M z^{-M}} \qquad (4.6)$$

Sec. 4.3 Cont. - 4

In this form, the IZT, x(n), may be obtained using one of the following methods:

* power series expansion method,
* partial fraction expansion method and
* residue method.

Sec. 4.3 Cont. - 5

Note:

- The z-transforms of common sequences are available, in closed form, and are given in the form of tables, such as **Table 4.1**.

- Such tables are useful in finding the inverse z-transform.

Sec. 4.3 Cont. - 6

Table 4.1 z-transforms .

Entry number	*Discrete-time sequence x(n), n ≥ 0*	*z-transform X(z)*	*Region of convergence of X(z)*
1	$k\delta(n)$	k	Everywhere
2	K	$\frac{kz}{z-1}$	|z| > 1
3	kn	$\frac{kz}{(z-1)^2}$	|z| > 1
4	kn^2	$\frac{kz(z+1)}{(z-1)^3}$	|z| > 1
5	$Ke^{-\alpha n}$	$\frac{kz}{z-e^{-\alpha}}$	|z| > $e^{-\alpha}$
6	$Kne^{-\alpha n}$	$\frac{kze^{-\alpha}}{(z-e^{-\alpha})^2}$	|z| > $e^{-\alpha}$
7	$1-e^{-\alpha n}$	$\frac{z(1-e^{-\alpha})}{z^2-z(1+e^{-\alpha})+e^{-\alpha}}$	|z| > $e^{-\alpha}$
8	$\cos(\alpha n)$	$\frac{z(z-\cos\alpha)}{z^2-2z\cos\alpha+1}$	|z| > 1

k and are constants; c is a complex number.

Sec. 4.3 Cont. - 7

Table 4.1 z-transforms - **Cont.**

9	$\sin(\alpha n)$	$\frac{z\sin\alpha}{z^2-2z\cos\alpha+1}$	$\lvert z\rvert > 1$
10	$E^{-\alpha n}\sin(\alpha n)$	$\frac{ze^{-\alpha}\sin\alpha}{z^2-2e^{-\alpha}z\cos\alpha+e^{-2\alpha}}$	$\lvert z\rvert > e^{-\alpha}$
11	$e^{-\alpha n}\cos(\alpha n)$	$\frac{ze^{-\alpha}(ze^{\alpha}-\cos\alpha)}{z^2-2ze^{-\alpha}\cos\alpha+e^{-2\alpha}}$	$\lvert z\rvert > e^{-\alpha}$
12	$\cosh(\alpha n)$	$\frac{z^2-z\cosh\alpha}{z^2-2z\cosh\alpha+1}$	$\lvert z\rvert > \cosh\alpha$
13	$\sinh(\alpha n)$	$\frac{z\sinh\alpha}{z^2-2z\cosh\alpha+1}$	$\lvert z\rvert > \sinh\alpha$
14	$k\alpha^n$	$\frac{kz}{z-\alpha}$	$\lvert z\rvert > \alpha$
15	$kn\alpha^n$	$\frac{k\alpha z}{(z-\alpha)^2}$	$\lvert z\rvert > \alpha$
16	$2\lvert c\rvert\,\lvert p\rvert^n\cos(n\angle p+\angle c)$	$\frac{cz}{z-p}+\frac{c^*z}{z-p^*}$	

k and are constants; c is a complex number.

Sec. 4.3.1

4.3.1 Power series method

Given the z-transform, X(z), of a causal sequence as in Eq. 4.6, it can be expanded into an infinite series in z^{-1} or z by long division (sometimes called synthetic division) :

$$X(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2} + \ldots + a_N z^{-N}}{b_0 + b_1 z^{-1} + b_2 z^{-2} + \ldots + b_M z^{-M}}$$

$$= x(0) + x(1) z^{-1} + x(2) z^{-2} + x(3) z^{-3} + \ldots \qquad (4.7)$$

In this method, the numerator and denominator of X(z) are first expressed in either descending powers of z or ascending powers of z –1 and the quotient is then obtained by long division. As illustrated in the following example.

Ch. 4 - 4.3.1 Cont. - 1

Example 4.2

Given the following z-transform of a causal LTI system, obtain its IZT by expanding it into a power series using long division:

$$X(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - z^{-1} + 0.3561z^{-2}}$$

Solution

First, we expand X(z) into a power series with the numerator and denominator polynomials in ascending power of z –1 and then perform the usual long division.

Ch. 4 - 4.3.1 Cont. - 2

Example 4.2 - Cont - 1

$$1 - z^{-1} + 0.3561z^{-2} \;\Big|\; \frac{1 + 3z^{-1} + 3.6439z^{-2} + 2.5756z^{-3} + \ldots}{1 + 2z^{-1} + z^{-2}}$$

$$\frac{1 - z^{-1} + 0.3561\, z^{-2}}{3z^{-1} + 0.6439\; z^{-2}}$$

$$\frac{3z^{-1} - 3z^{-2} + 1.0683z^{-3}}{3.6439z^{-2} - 1.0683z^{-3}}$$

$$\frac{3.6439z^{-2} - 3.6439z^{-3} + 1.2975927z^{-4}}{2.5756z^{-3} - 1.2975927z^{-4}}$$

Ch. 4 - 4.3.1 Cont. - 3

Example 4.2 - Cont - 2

Alternatively, we may express the numerator and denominator in positive powers of z, in descending order, and then perform the long division:

$$\begin{array}{r} \dfrac{z^2+2z+1}{z^2-z+0.3561} \\ \dfrac{1+3z^{-1}+3.6439z^{-2}+2.5756z^{-3}+\ldots}{z^2+2z+1} \\ \underline{z^2-z+0.3561} \\ \underline{z^2-z+0.3561} \end{array}$$

$$\begin{array}{l} 3z+0.6439 \\ 3z-3+1.0683\ z^{-1} \\ \hline 3.6439-3.64391\ z^{-1}+1.2975927\ z^{-2} \\ 2.5756\ z^{-1}-1.2975927\ z^{-2} \end{array}$$

Ch. 4 - 4.3.1 Cont. - 4

Example 4.2 - Cont - 3

Either way, the z-transform is now expanded into the familiar power series, that is

$$X(z) = 1+3z^{-1} + 3.6439z^{-2} + 2.576z^{-3} + \ldots$$

The inverse z-transform (x(n)) can now be written down directly :

$$x(0) = 1;\ x(1) = 3;\ x(2) = 3.6439;\ x(3) = 2.5756;\ \ldots\ldots$$

Sec. 4.3.1 Cont. - 5

The long division approach can be reformulated so that the values of x(n) are obtained recursively as :

$$x(n) = \left[a_n - \sum_{i=1}^{n} x(n-i) b_i \right] / b_0, \; n = 1, 2, \ldots \qquad (4.8a)$$

where

$$x(0) = a_0 / b_0 \qquad (4.8b)$$

Ch. 4 - 4.3.1 Cont. - 6

Example 4.3

We Find the first four terms of the inverse z-transform, x(n), using the **recursive approach**. Assume that the z-transform, X(z), is the same as in **Example 4.2**, that is repeat the previous example to illustrate the recursive approach.

$$X(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - z^{-1} + 0.3561z^{-2}}$$

Solution

Comparing the coefficients of X(z) above with those of the general transform in Equation 4.7 we have

$$a_0 = 1,\ a_1 = 2,\ a_2 = 1,\ b_0 = 1,\ b_1 = -1,\ b_2 = 0.3561;\ N = M = 2$$

Ch. 4 - 4.3.1 Cont. - 7

Example 4.3 - Cont - 1

From Equation 4.8 we have

$x(0) = a0/b0 = 1$
$x(1) = [a1 – x(0)b1]/b0 = [2 – 1\ (-1)] = 3$
$x(2) = [a2 – x(1)b1– x(0)b2] = 1 – 3\ (-1) – 1\ \ 0.3561 = 3.6439$
$x(3) = [a3 – x(2)b1 – x(1)b2 + x(0)b3]$
$= 0 – x(2)b1 – x(1)b2 = 0 – 3.6439\ \ (-1) – 3*0.3561 = 2.5756$

Thus the first four values of the inverse z-transform are:

$$x(0) = 1,\ x(1) = 3,\ x(2) = 3.6439,\ x(3) = 2.5756$$

It is seen that both the recursive and direct, long division methods lead to identical solutions.